

Abnormal Adult Physiological Values in Emergency Cardiac & Respiratory Conditions

This document summarizes abnormal physiological thresholds commonly used to identify life-threatening cardiac and respiratory conditions in adults. Values apply to resting, nonpregnant adults. Reference numbers are indicated in brackets.

Condition	Parameter	Abnormal Value	Reference
Tachycardia	Heart Rate	>100 beats/min	[1,2]
Bradycardia	Heart Rate	<60 beats/min	[1,2]
Respiratory Distress	Respiratory Rate	>20 or <12 breaths/min	[2,3]
Hypothermia	Core Temperature	<35 °C	[4]
Hypertensive Crisis	Blood Pressure	≥180/120 mmHg	[1,3]
Atrial Fibrillation (Suspected)	Rhythm	Irregularly irregular, absent P waves	[1,2]
Atrial Fibrillation	Heart Rate	Often >100 beats/min	[1]
Myocardial Infarction (Suspected)	Troponin I/T	>99th percentile (lab specific)	[1,2]
Myocardial Infarction (Suspected)	ECG	ST elevation/depression, T inversion	[1,2]
Hypotension/Shock	Systolic BP	<90 mmHg	[3]
Poor Perfusion	MAP	<65 mmHg	[3]
Hypoxemia	SpO ₂	<90%	[1,5]

References

1. American Heart Association (AHA). ACLS Guidelines and ECG Interpretation Standards.
2. Harrison's Principles of Internal Medicine, 21st Edition.
3. Davidson's Principles and Practice of Medicine, 24th Edition.
4. World Health Organization (WHO). Hypothermia Definition and Classification.
5. UpToDate®. Evaluation of Hypoxemia and Respiratory Distress.